

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q18-0968

Atoka 138 kV
September 02, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0968. The Interconnection Customer proposes a 46.6 MW solar photovoltaic facility interconnecting at the Atoka 138 kV substation in the Virginia area. The request was received on March 27, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 358 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,784,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Claytor / Bladenboro Solar 2 138.0 kV	Lowmoor / Hinton Ap 345.0 kV line	113.6	230	5.3
S Princeton / Bandhead 138.0 kV line	New Hope Ap / Manatee 345.0 kV	113.4	295	6.6
Pitchgen / Lochmere Hydroelectric Plant 138.0 kV line	Manatee Substation / Hinton Ap 345.0 kV	112.2	173	6.2
New Hope Ap 138/1 kV Autotransformer	New Hope Ap Substation / Manatee 345.0 kV line	111.9	313	6.6
Bandhead / New Hope Ap 138.0 kV	Manatee / New Hope Ap 345.0 kV	111.8	313	6.6
Hartman Creek 138 kV/1 kV Autotransformer	Lowmoor / Hinton Ap 345.0 kV	111.0	384	8.3
Turkey Pen / Garden Crk 138.0 kV	Carlton Hills / Clinch Riv Substation 345.0 kV	110.3	326	7.3
Friendship / Vacherie 138.0 kV	Kean Mtn And Grassy Crk Substation / Barr Lake 345.0 kV line	109.7	273	5.5
Parke 345-1 kV Transformer	Barr Lake / New Hope Ap 345.0 kV	109.1	438	10.4
Mcadenville / Vacherie 138.0 kV Line	Barr Lake / Kean Mtn And Grassy Crk 345.0 kV	108.6	256	5.5
Sringbank / Station 11 138.0 kV L	Barr Lake / Kean Mtn And Grassy Creek 345.0 kV L	108.6	318	7.7
Stock Island / Station 11 138.0 kV	Kean Mtn And Grassy Crk / Barr Lake 345.0 kV	107.4	337	7.7
Shullsburg / Trezevat 138.0 kV	Lowmoor / Hinton Ap 345.0 kV	105.9	362	6.2

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Trezevant / Rogue 138.0 kV	Lowmoor / Hinton Ap 345.0 kV	104.6	388	6.2
Lynn / Rogue 138.0 kV line	Lowmoor Substation / Hinton Ap 345.0 kV L	104.5	392	6.2
Lynn / Lost Nation 138.0 kV	Lowmoor / Hinton Ap 345.0 kV	104.2	398	6.2
Lost Nation / No Troy 138.0 kV	Hinton Ap / Lowmoor 345.0 kV	104.0	402	6.2
Briar / Pitchgen 138.0 kV	Manatee / Hinton Ap 345.0 kV Line	103.6	290	9.2
Fraser / Briar 138.0 kV Line	Hinton Ap / Manatee 345.0 kV	103.5	293	9.2
Hinton Ap / Lowmoor 345.0 kV	Vaseaux Lake / McClain Energy Facility Substation 138.0 kV	103.4	1,433	20.6
Garden Creek / Kean Mtn And Grassy Creek Substation 138.0 kV	Carlton Hills / Clinch Riv 345.0 kV	103.3	338	7.3
New Hope Ap / Manatee 345.0 kV	Parke / Barr Lake 345.0 kV Line	102.7	1,153	23.1
Turkey Pen / Leesburg Picciola 138.0 kV L	Clinch River / Carlton Hills 345.0 kV L	102.0	235	5.6
Ou Spirit Wind Farm / Fraser 138.0 kV	Manatee / Hinton Ap 345.0 kV	100.9	264	7.1
Clinch River Substation / Carlton Hills 345.0 kV	Kean Mtn And Grassy Creek 138/1 kV Auto	100.8	1,154	28.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Claytor / Bladenboro Solar 2 138.0 kV	Rebuild line with higher-capacity conductor	\$1,046,000	3% (\$30,000)
Bandhead / S Princeton 138.0 kV	Rebuild line with higher-capacity conductor	\$7,157,000	2% (\$179,000)
Lochmere Hydroelectric Plant / Pitchgen 138.0 kV	Rebuild line with higher-capacity conductor	\$6,191,000	3% (\$161,000)
New Hope Ap 138/1 kV Autotransformer	Replace transformer with higher-rated unit	\$3,411,000	2% (\$85,000)
Bandhead / New Hope Ap 138.0 kV	Rebuild line with higher-capacity conductor	\$1,974,000	2% (\$49,000)
Hartman Creek 138 kV/1 kV Autotransformer	Replace transformer with higher-rated unit	\$5,215,000	2% (\$132,000)
Garden Crk / Turkey Pen 138.0 kV	Rebuild line with higher-capacity conductor	\$6,464,000	3% (\$195,000)
Friendship / Vacherie 138.0 kV	Reconductor line	\$1,666,000	3% (\$54,000)
Parke Substation 345/1 kV Autotransformer	Replace transformer with higher-rated unit	\$16,127,000	4% (\$588,000)
Mcadenville / Vacherie 138.0 kV	Reconductor line	\$8,450,000	3% (\$274,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Springbank / Sta 11 138.0 kV Line	Reconductor line	\$5,239,000	3% (\$134,000)
Stock Island / Station 11 138.0 kV	Reconductor line	\$1,727,000	3% (\$44,000)
Trezevant / Shullsburg 138.0 kV	Reconductor line	\$641,000	3% (\$16,000)
Trezevant / Rogue 138.0 kV	Reconductor line	\$7,882,000	3% (\$202,000)
Lynn / Rogue 138.0 kV line	Reconductor line	\$3,417,000	3% (\$88,000)
Lynn / Lost Nation 138.0 kV	Reconductor line	\$2,536,000	3% (\$65,000)
Lost Nation / No Troy 138.0 kV	Reconductor line	\$1,707,000	3% (\$44,000)
Briar / Pitchgen 138.0 kV	Reconductor line	\$1,213,000	2% (\$30,000)
Fraser / Briar 138.0 kV Line	Reconductor line	\$665,000	2% (\$17,000)
Hinton Ap / Lowmoor 345.0 kV	Reconductor line	\$67,314,000	1% (\$591,000)
Garden Creek / Kean Mtn And Grassy Creek Substation 138.0 kV	Reconductor line	\$2,197,000	3% (\$66,000)
New Hope Ap / Manatee 345.0 kV Line	Reconductor line	\$11,609,000	2% (\$171,000)
Turkey Pen / Leesburg Picciola 138.0 kV L	Reconductor line	\$10,781,000	4% (\$409,000)
Ou Spirit Wind Farm / Fraser 138.0 kV	Reconductor line	\$2,695,000	2% (\$66,000)
Carlton Hills / Clinch Riv 345.0 kV	Reconductor line	\$5,200,000	2% (\$94,000)
	Total allocated to project		\$3,784,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.